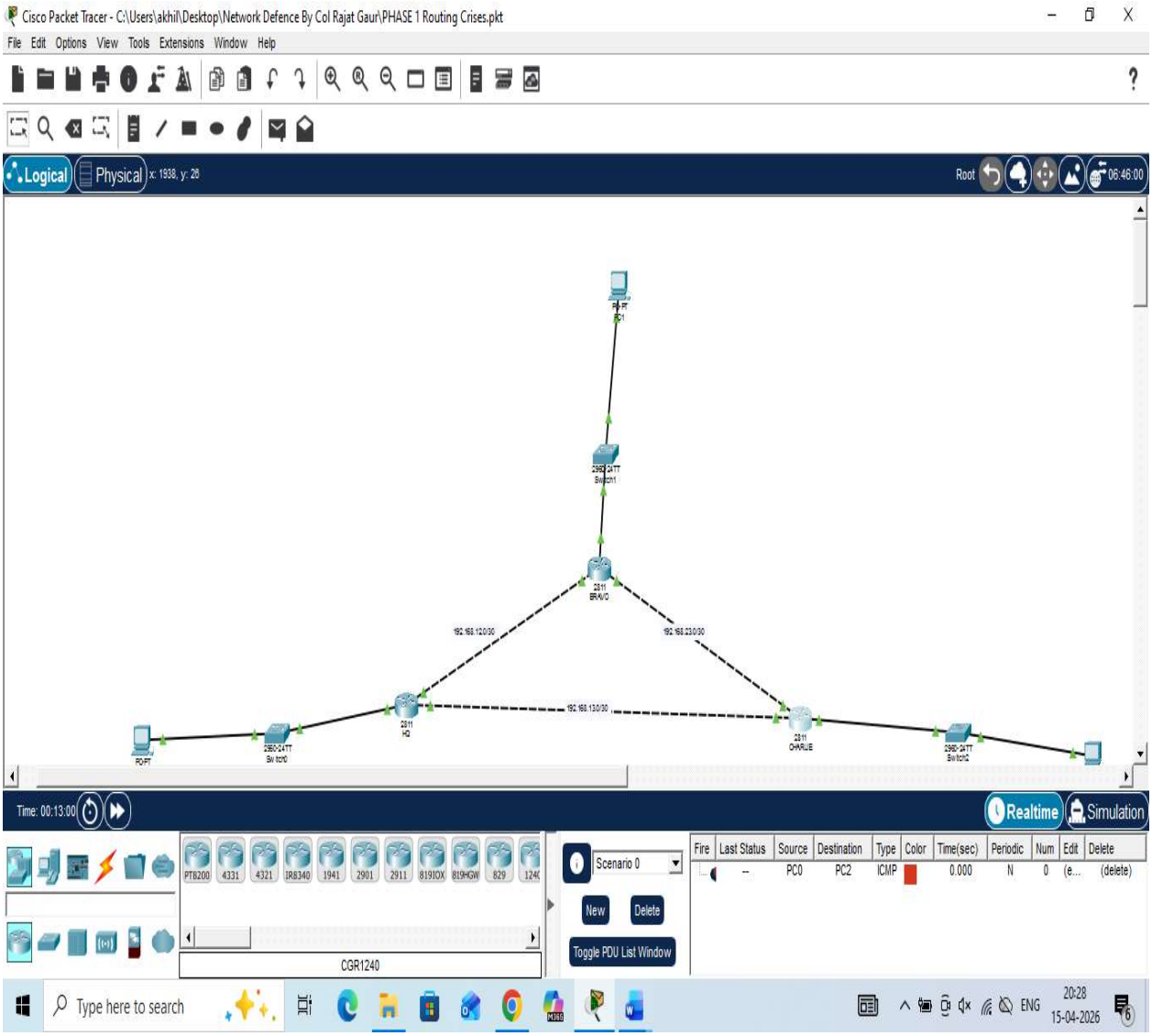
Phase 1 – Routing Crisis (Cisco Packet Tracer)

Situation

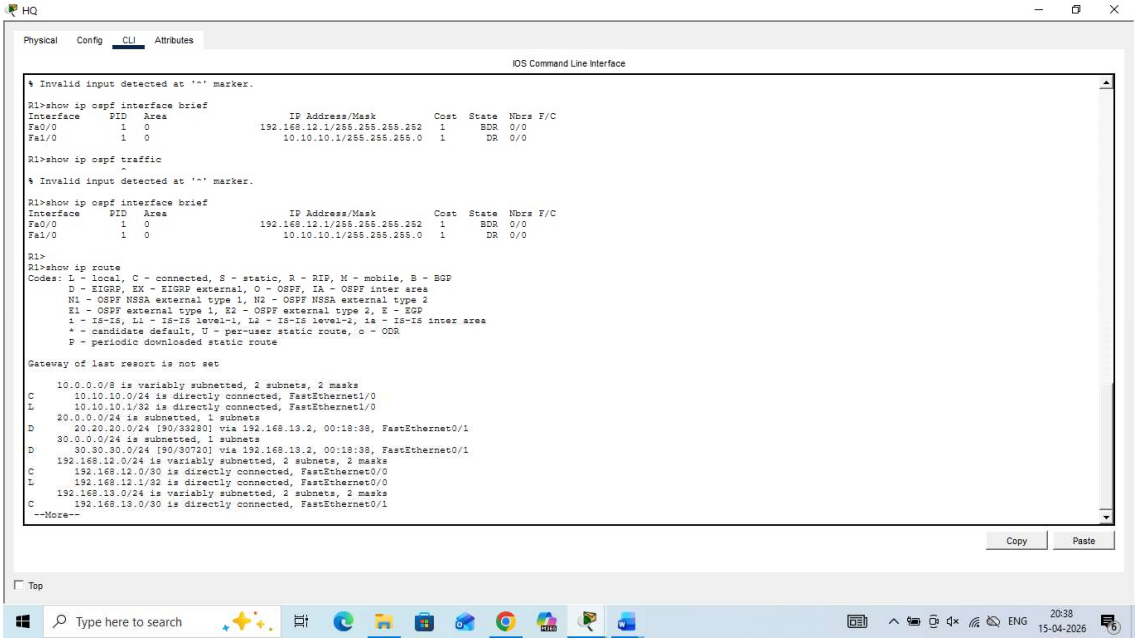
HQ reports that Bravo can ping Charlie, but HQ cannot reach Charlie.

Tasks

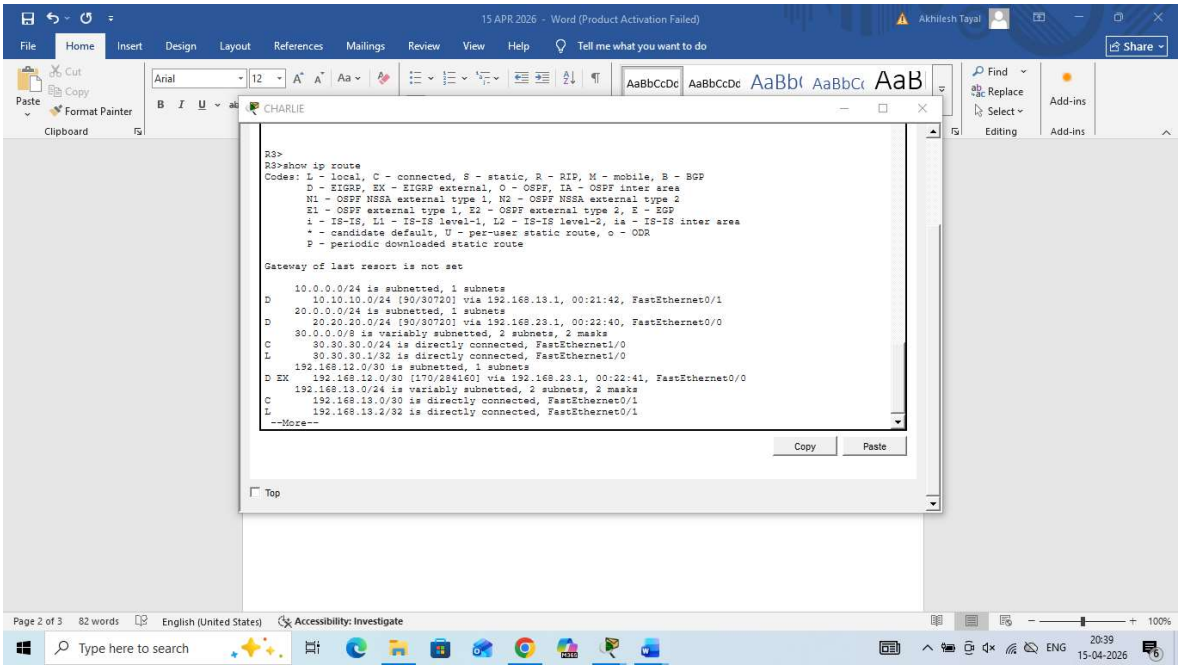
1. Build the above topology in Cisco Packet Tracer.
2. Configure:
    - R1 and R2 using **OSPF**
    - R3 using **EIGRP**



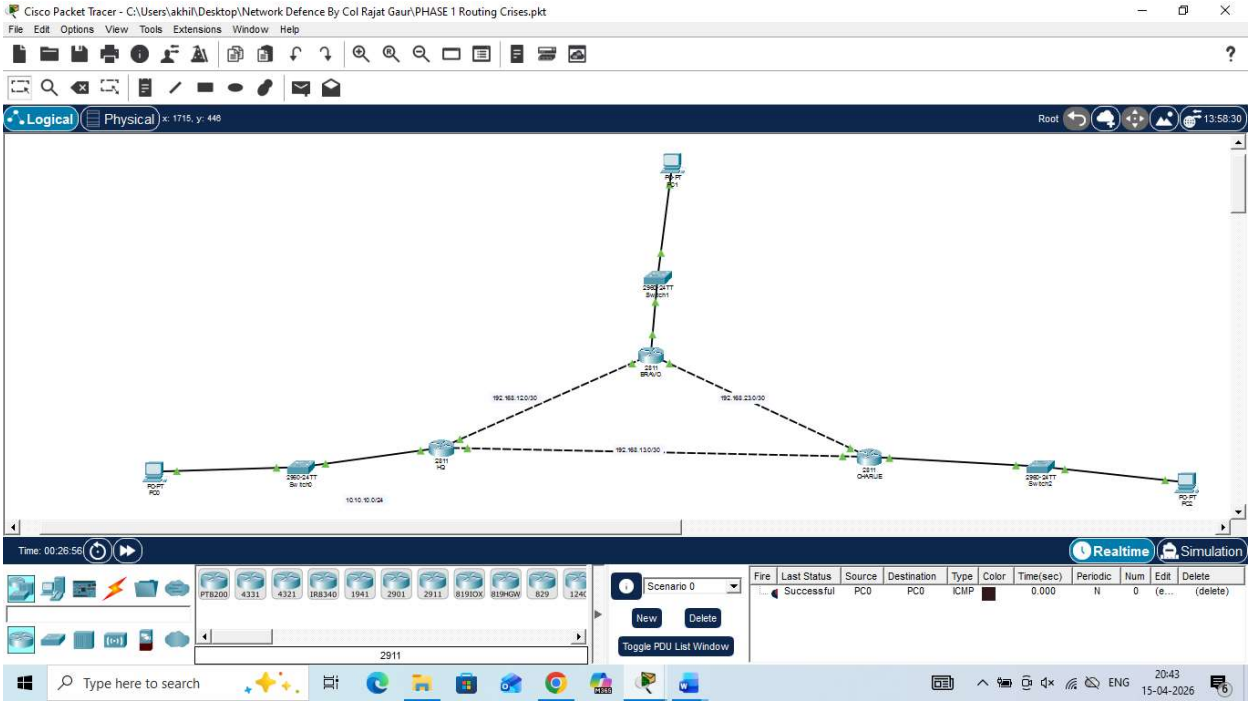
R1 and R2 using **OSPF**



R3 using **EIGRP**

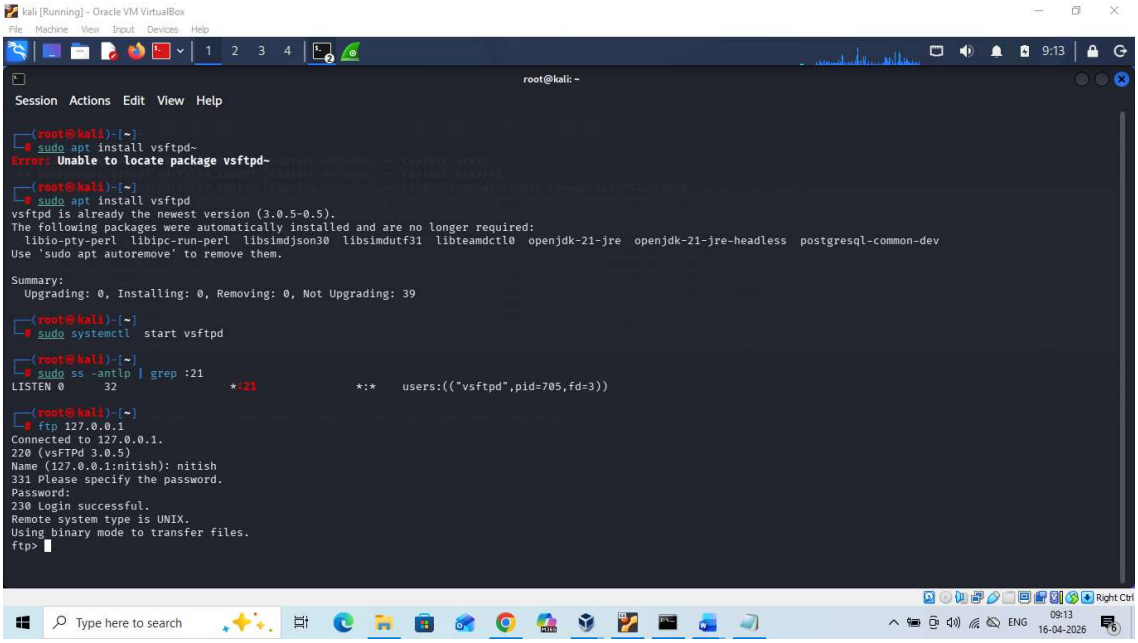


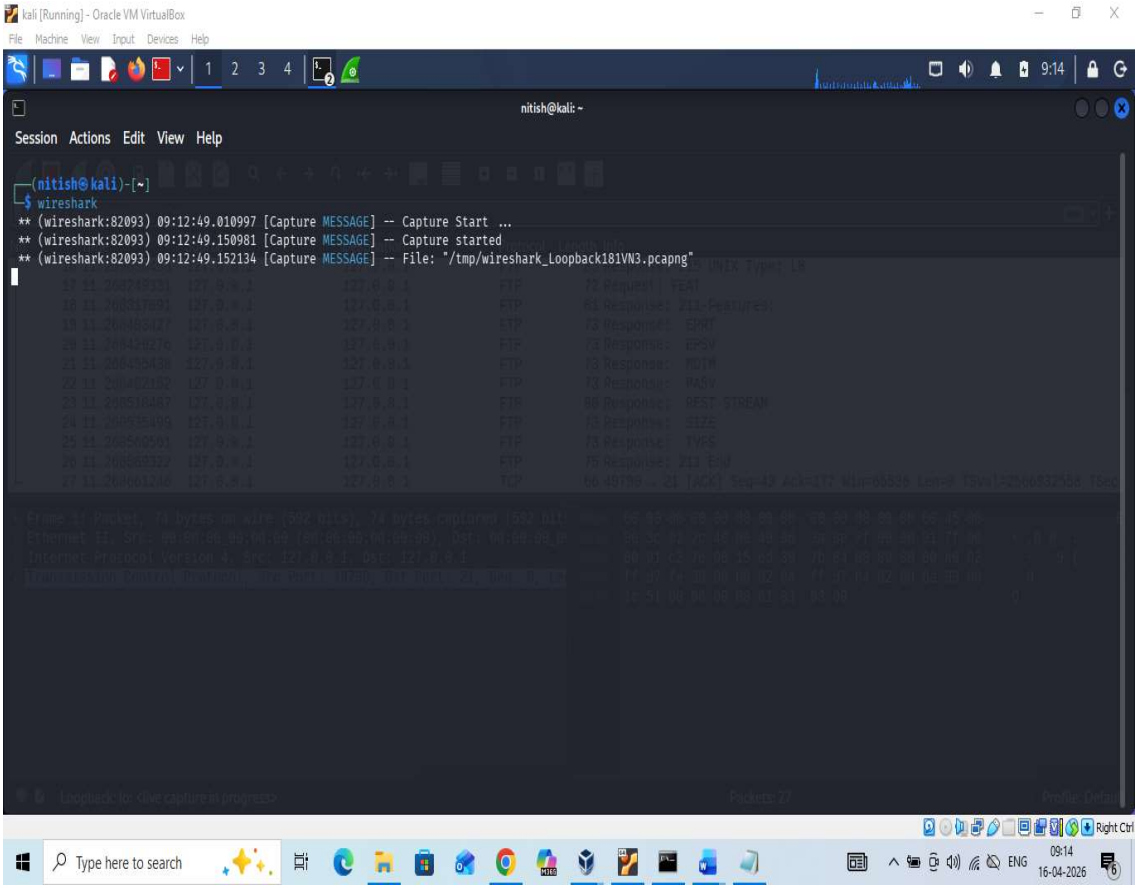
Redistribute routes where required.

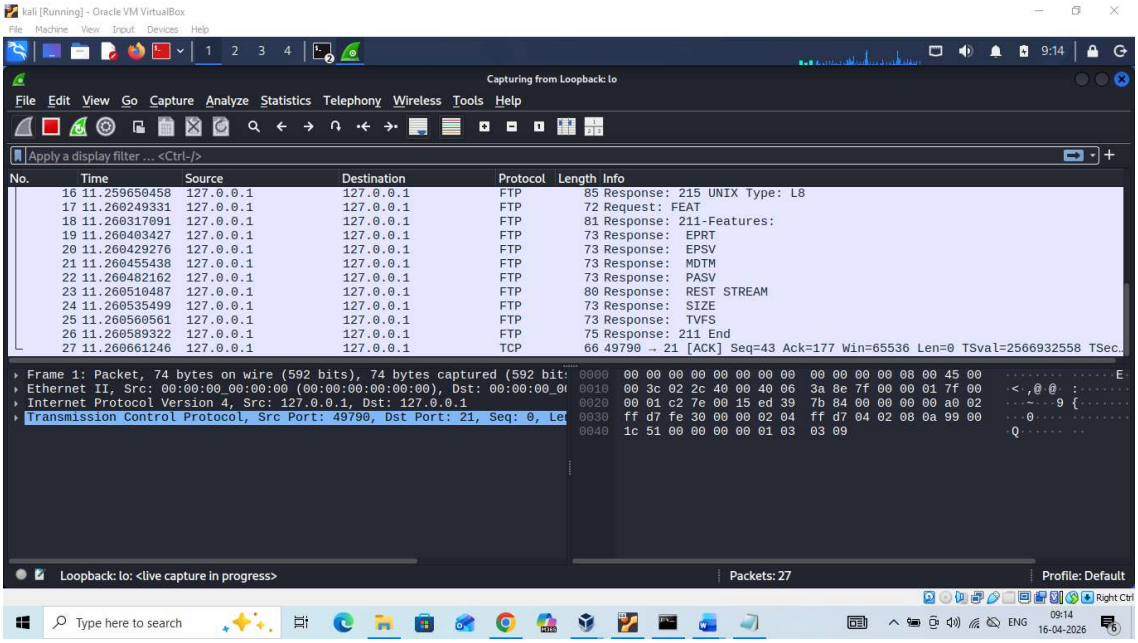


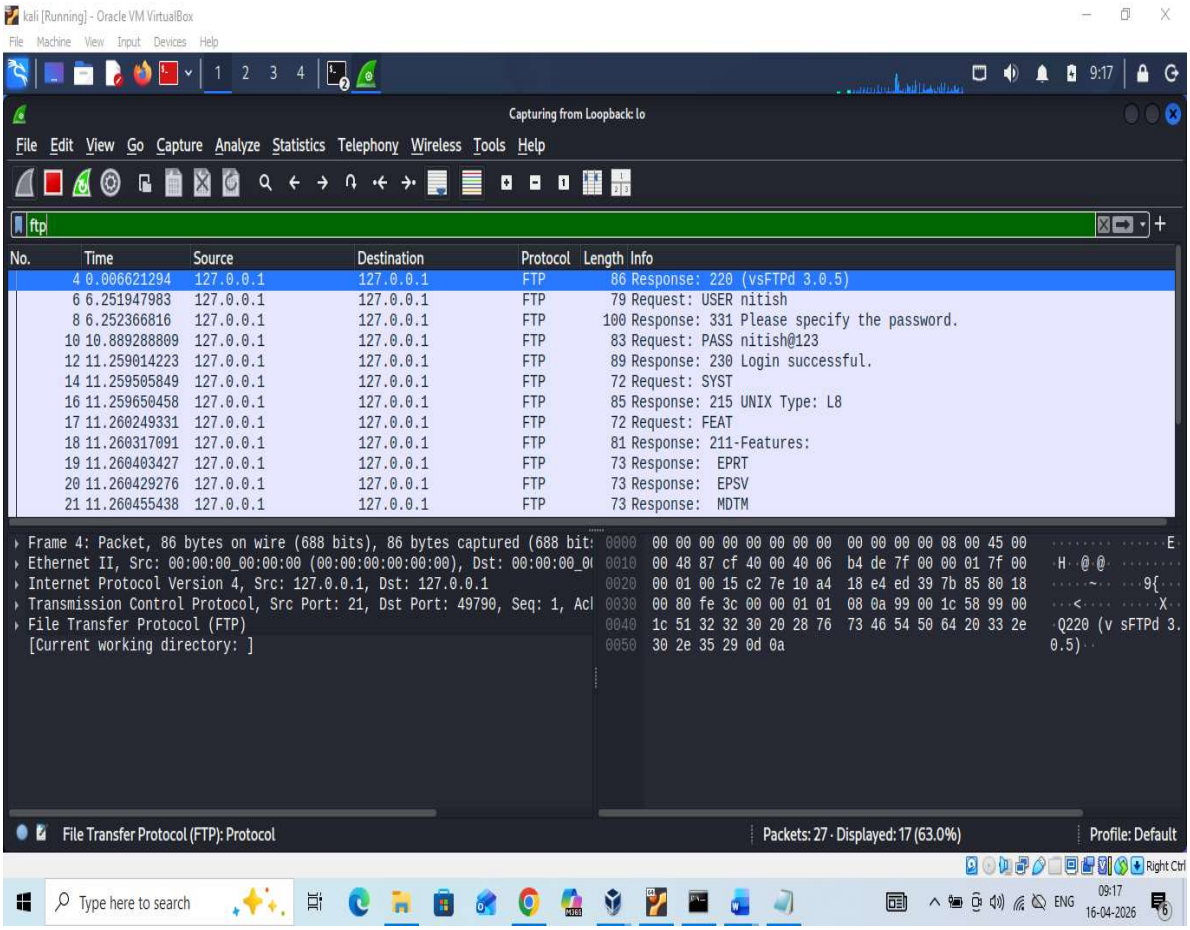
Answer No 2 :

FTP Login

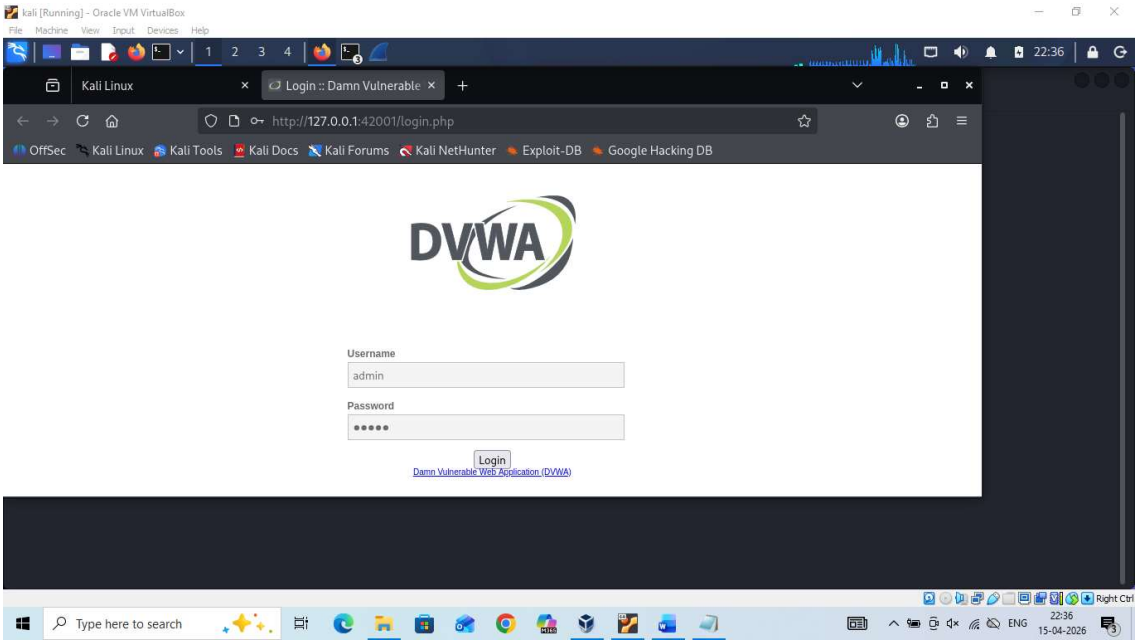


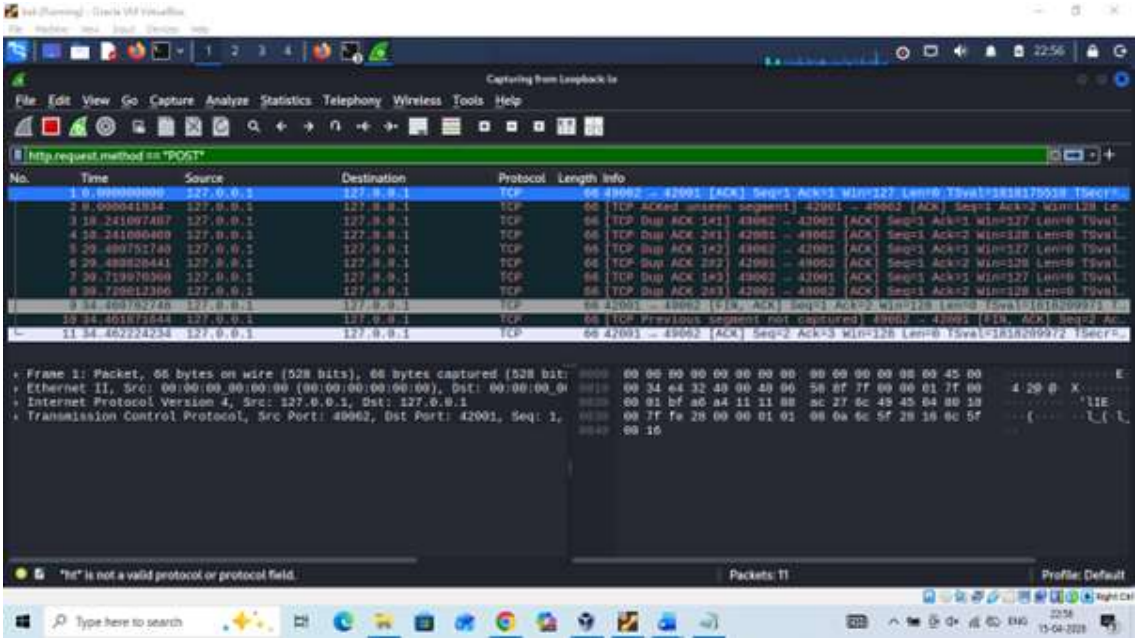


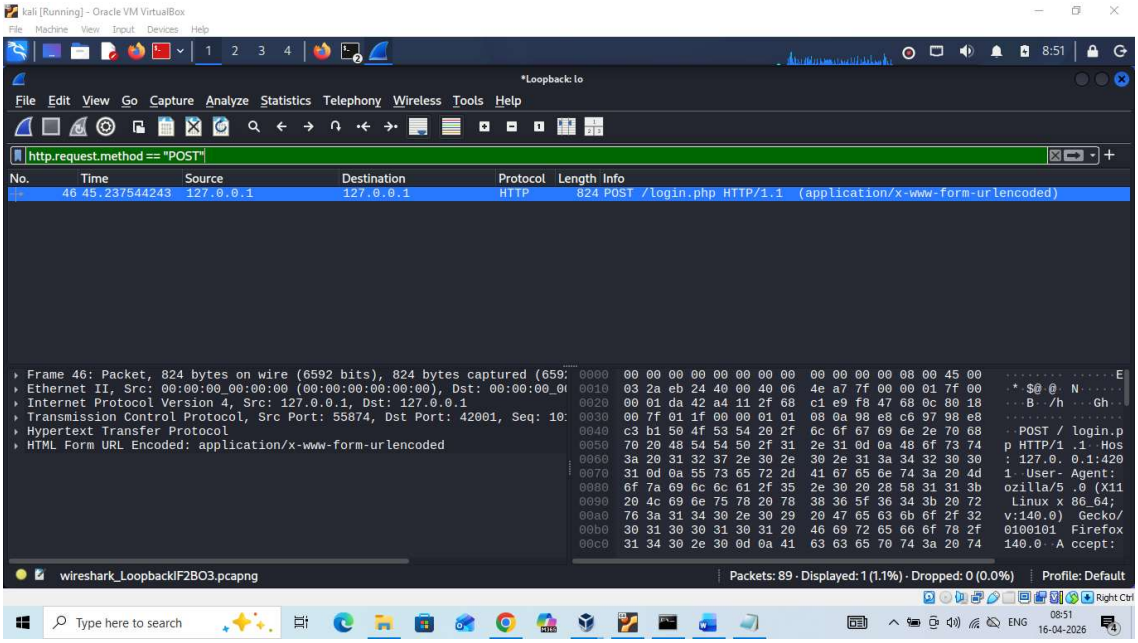

kali [Running] - Oracle VM VirtualBox
Capturing from Loopback: lo
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
16 11.259650458 127.0.0.1 127.0.0.1 FTP 85 Response: 215 UNIX Type: L8
17 11.260249331 127.0.0.1 127.0.0.1 FTP 72 Request: FEAT
18 11.260317091 127.0.0.1 127.0.0.1 FTP 81 Response: 211-Features:
19 11.260403427 127.0.0.1 127.0.0.1 FTP 73 Response: EPRT
20 11.260429276 127.0.0.1 127.0.0.1 FTP 73 Response: EPSV
21 11.260455438 127.0.0.1 127.0.0.1 FTP 73 Response: MDTM
22 11.260482162 127.0.0.1 127.0.0.1 FTP 73 Response: PASV
23 11.260510487 127.0.0.1 127.0.0.1 FTP 80 Response: REST STREAM
24 11.260535499 127.0.0.1 127.0.0.1 FTP 73 Response: SIZE
25 11.260560561 127.0.0.1 127.0.0.1 FTP 73 Response: TVFS
26 11.260589322 127.0.0.1 127.0.0.1 FTP 75 Response: 211 End
27 11.260661246 127.0.0.1 127.0.0.1 TCP 66 49790 → 21 [ACK] Seq=43 Ack=177 Win=65536 Len=0 TSval=2566932558 TSec
Frame 1: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bit
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_0
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 49790, Dst Port: 21, Seq: 0, Le
0000 00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00
0010 00 3c 02 2c 40 00 40 06 3a 8e 7f 00 00 01 7f 00
0020 00 01 c2 7e 00 15 ed 39 7b 84 00 00 00 00 a0 02
0030 ff d7 fe 30 00 00 02 04 ff d7 04 02 08 0a 99 00
0040 1c 51 00 00 00 00 01 03 03 09
Loopback: lo: <live capture in progress>
Packets: 27
Profile: Default
Type here to search
ENG
09:14
16-04-2026

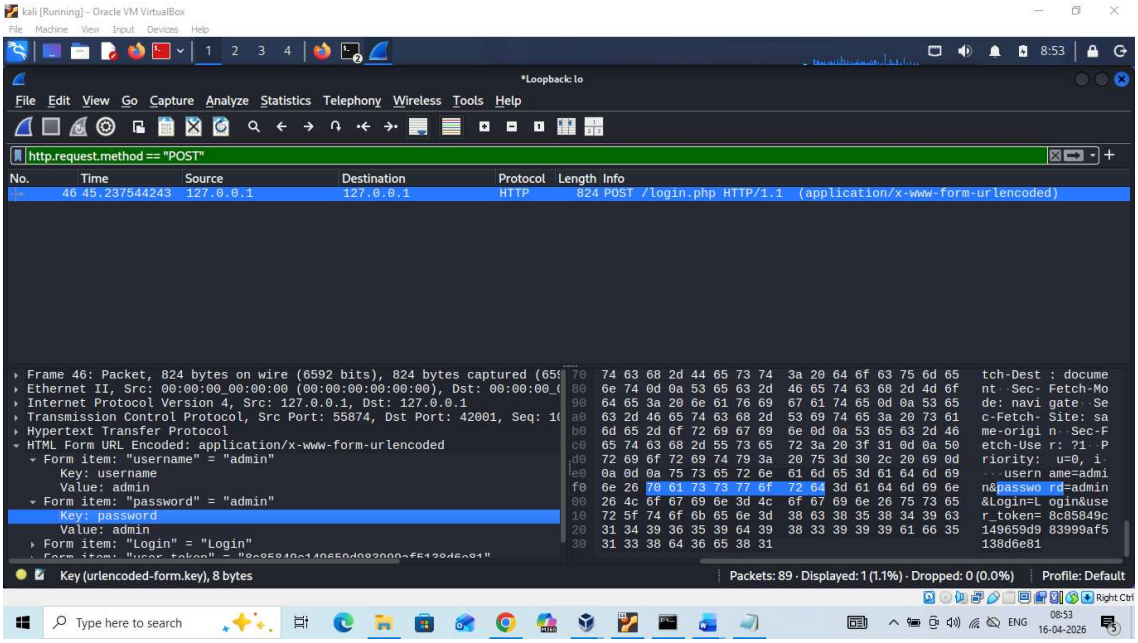

kali [Running] - Oracle VM VirtualBox
Capturing from Loopback: lo
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ftp
No. Time Source Destination Protocol Length Info
4 0.006621294 127.0.0.1 127.0.0.1 FTP 86 Response: 220 (vsFTPd 3.0.5)
6 6.251947983 127.0.0.1 127.0.0.1 FTP 79 Request: USER nitish
8 6.252366816 127.0.0.1 127.0.0.1 FTP 100 Response: 331 Please specify the password.
10 10.889288809 127.0.0.1 127.0.0.1 FTP 83 Request: PASS nitish@123
12 11.259014223 127.0.0.1 127.0.0.1 FTP 89 Response: 230 Login successful.
14 11.259505849 127.0.0.1 127.0.0.1 FTP 72 Request: SYST
16 11.259650458 127.0.0.1 127.0.0.1 FTP 85 Response: 215 UNIX Type: L8
17 11.260249331 127.0.0.1 127.0.0.1 FTP 72 Request: FEAT
18 11.260317091 127.0.0.1 127.0.0.1 FTP 81 Response: 211-Features:
19 11.260403427 127.0.0.1 127.0.0.1 FTP 73 Response: EPRT
20 11.260429276 127.0.0.1 127.0.0.1 FTP 73 Response: EPSV
21 11.260455438 127.0.0.1 127.0.0.1 FTP 73 Response: MDTM
Frame 4: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bit
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_0
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 21, Dst Port: 49790, Seq: 1, Ac
File Transfer Protocol (FTP)
[Current working directory:]
0000 00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00
0010 00 48 87 cf 40 00 40 06 b4 de 7f 00 00 01 7f 00
0020 00 01 00 15 c2 7e 10 a4 18 e4 ed 39 7b 85 80 18
0030 00 80 fe 3c 00 00 01 01 08 0a 99 00 1c 58 99 00
0040 1c 51 32 32 30 20 28 76 73 46 54 50 64 20 33 2e
0050 30 2e 35 29 0d 0a
Q220 (v sFTPd 3.
0.5)
File Transfer Protocol (FTP): Protocol
Packets: 27 · Displayed: 17 (63.0%)
Profile: Default
Type here to search
ENG
09:17
16-04-2026

DVWA

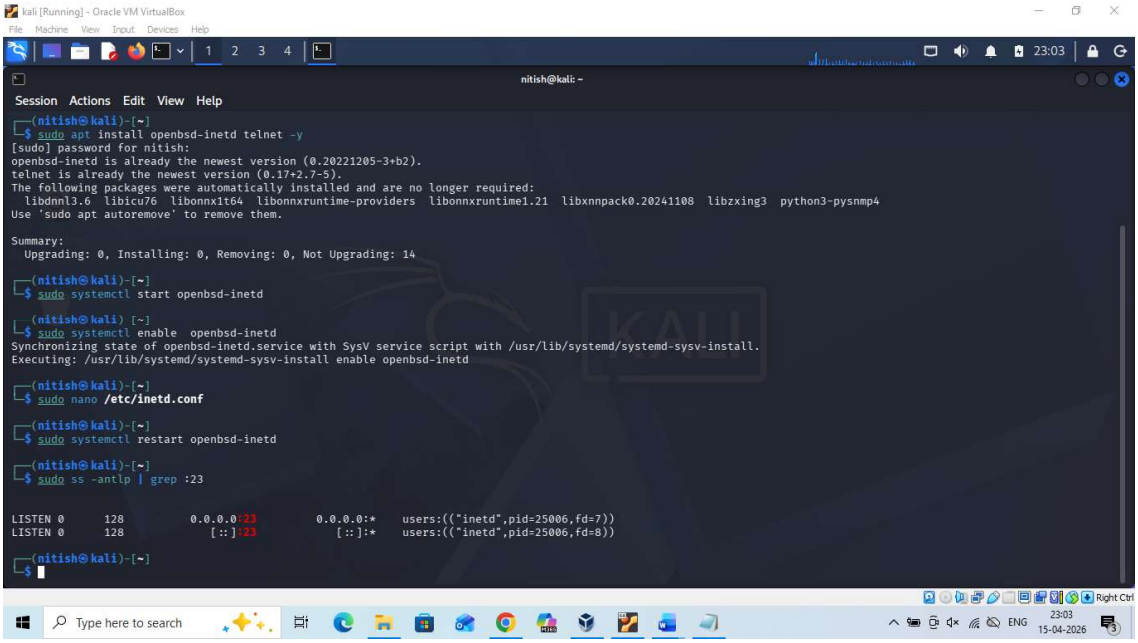

kali [Running] - Oracle VM VirtualBox
Kali Linux
Login :: Damn Vulnerable
http://127.0.0.1:42001/login.php
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
DVWA
Username
admin
Password
Login
Damn Vulnerable Web Application (DVWA)
Type here to search

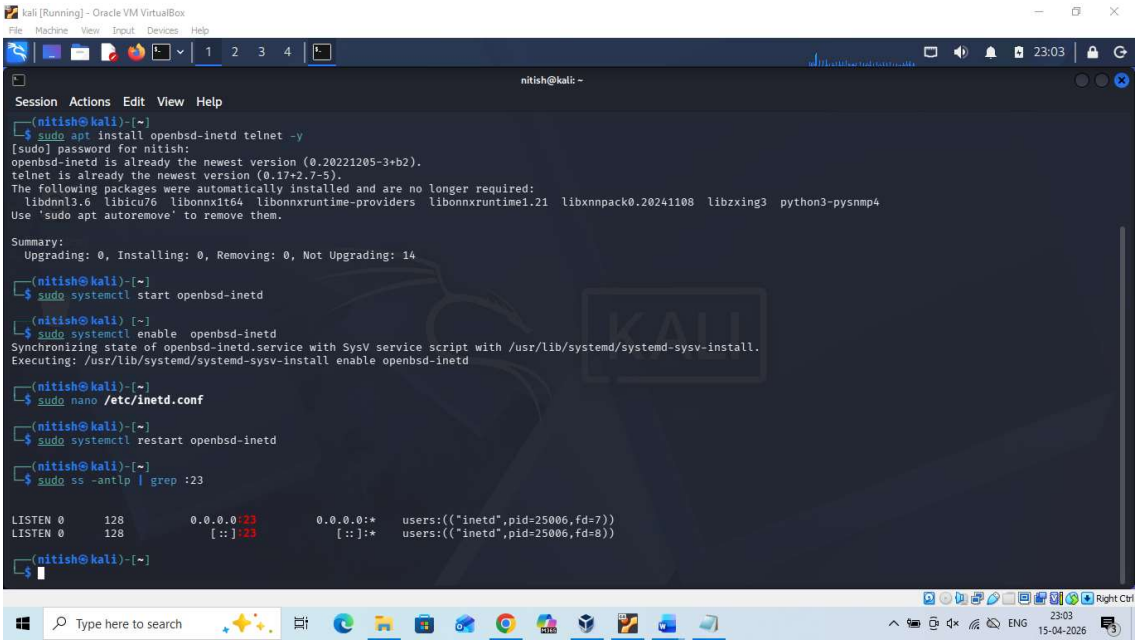

Capturing from Loopback: lo
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http.request.method == "POST"
No. Time Source Destination Protocol Length Info
Frame 1: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits)
Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 49062, Dst Port: 42001, Seq: 1,
"ht" is not a valid protocol or protocol field.
Packets: 11
Profile: Default
Type here to search

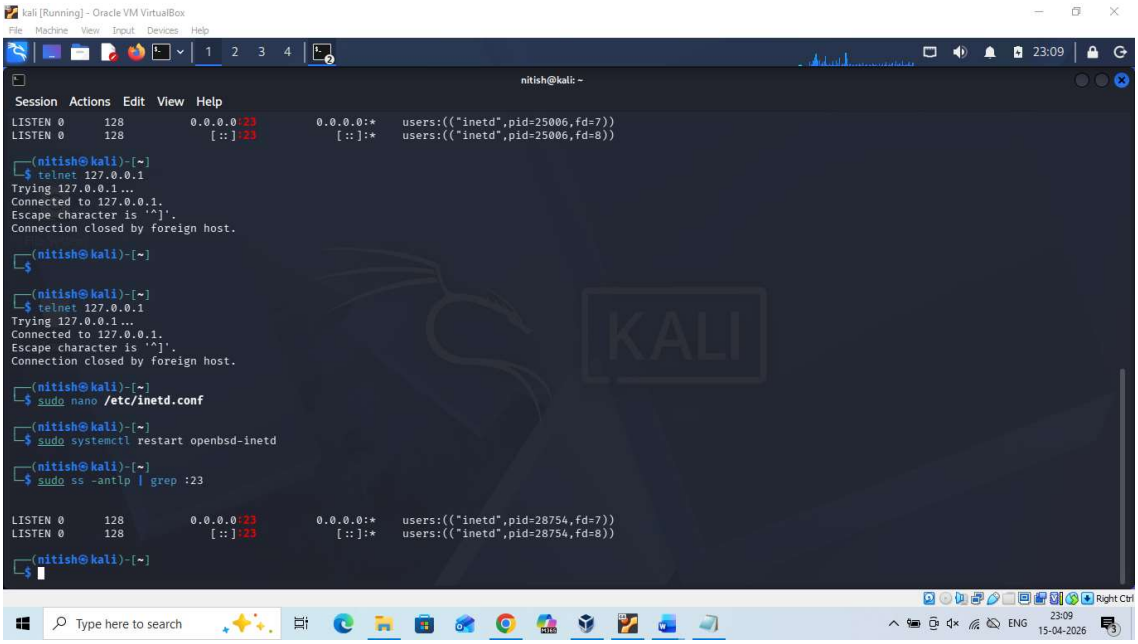

kali [Running] - Oracle VM VirtualBox
*Loopback: lo
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http.request.method == "POST"
No. Time Source Destination Protocol Length Info
46 45.237544243 127.0.0.1 127.0.0.1 HTTP 824 POST /login.php HTTP/1.1 (application/x-www-form-urlencoded)
Frame 46: Packet, 824 bytes on wire (6592 bits), 824 bytes captured (659
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 55874, Dst Port: 42001, Seq: 10
Hypertext Transfer Protocol
HTML Form URL Encoded: application/x-www-form-urlencoded
wireshark_LoopbackIF2BO3.pcapng
Packets: 89 · Displayed: 1 (1.1%) · Dropped: 0 (0.0%)
Profile: Default
Type here to search
ENG
08:51
16-04-2026

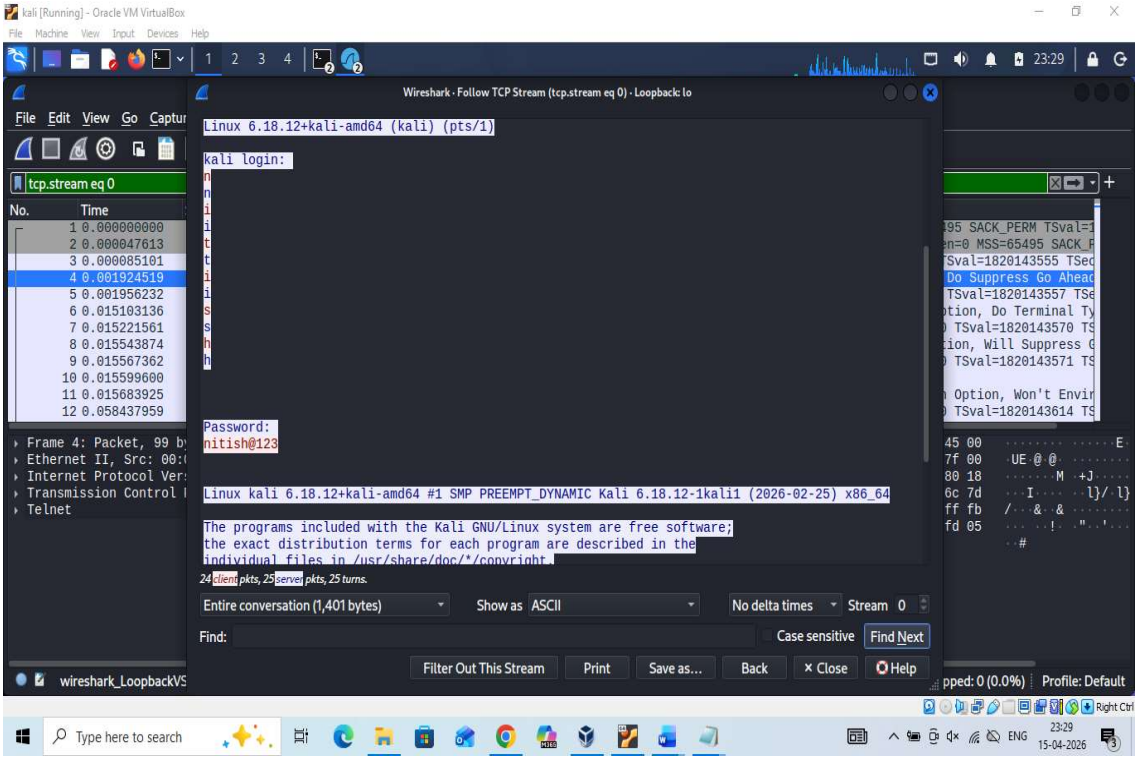

kali [Running] - Oracle VM VirtualBox
*Loopback: lo
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http.request.method == "POST"
No. Time Source Destination Protocol Length Info
46 45.237544243 127.0.0.1 127.0.0.1 HTTP 824 POST /login.php HTTP/1.1 (application/x-www-form-urlencoded)
Frame 46: Packet, 824 bytes on wire (6592 bits), 824 bytes captured (659
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_0
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 55874, Dst Port: 42001, Seq: 1
Hypertext Transfer Protocol
HTML Form URL Encoded: application/x-www-form-urlencoded
Form item: "username" = "admin"
Key: username
Value: admin
Form item: "password" = "admin"
Key: password
Value: admin
Form item: "Login" = "Login"
Key (urlencoded-form.key), 8 bytes
Packets: 89 · Displayed: 1 (1.1%) · Dropped: 0 (0.0%)
Profile: Default
Type here to search
ENG
08:53
16-04-2026

TELNET

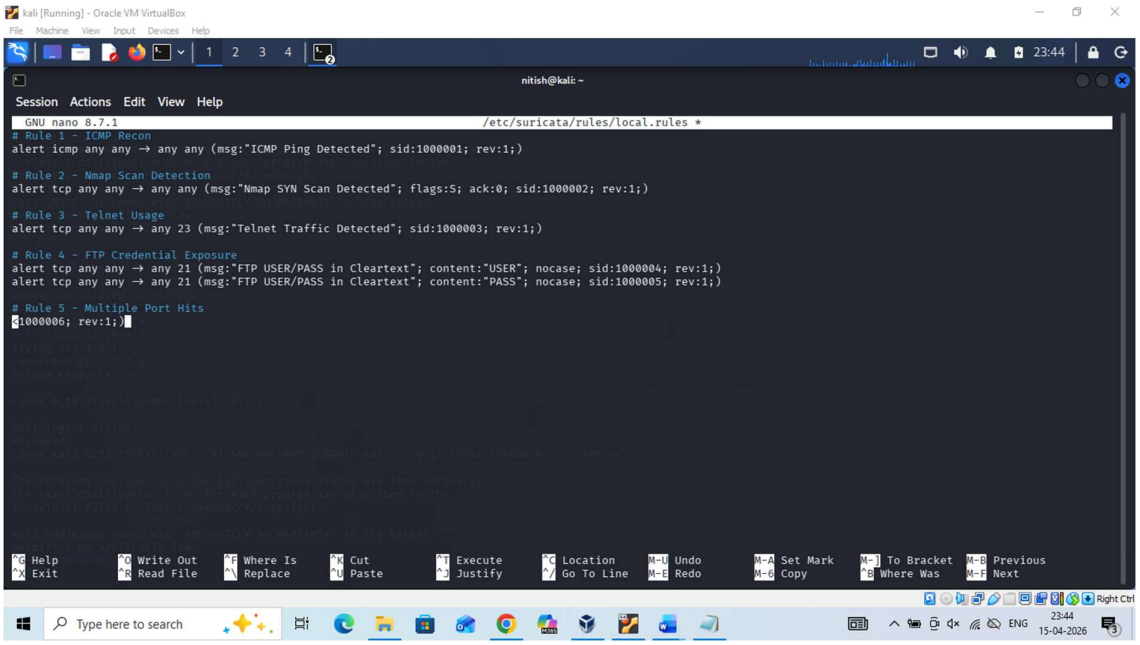

kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
nitish@kali: ~
Session Actions Edit View Help
┌──(nitish㉿kali)-[~]
└─$ sudo apt install openbsd-inetd telnet -y
[sudo] password for nitish:
openbsd-inetd is already the newest version (0.20221205-3+b2).
telnet is already the newest version (0.17+2.7-5).
The following packages were automatically installed and are no longer required:
libdnnl3.6 libicu76 libonnx1t64 libonnxruntime-providers libonnxruntime1.21 libxnnpack0.20241108 libzxing3 python3-pysnmp4
Use 'sudo apt autoremove' to remove them.
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 14
┌──(nitish㉿kali)-[~]
└─$ sudo systemctl start openbsd-inetd
┌──(nitish㉿kali)-[~]
└─$ sudo systemctl enable openbsd-inetd
Synchronizing state of openbsd-inetd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable openbsd-inetd
┌──(nitish㉿kali)-[~]
└─$ sudo nano /etc/inetd.conf
┌──(nitish㉿kali)-[~]
└─$ sudo systemctl restart openbsd-inetd
┌──(nitish㉿kali)-[~]
└─$ sudo ss -antlp | grep :23
LISTEN 0 128 0.0.0.0:23 0.0.0.0:* users:(("inetd",pid=25006,fd=7))
LISTEN 0 128 [::]:23 [::]:* users:(("inetd",pid=25006,fd=8))
┌──(nitish㉿kali)-[~]
└─$
Type here to search
ENG
23:03
15-04-2026

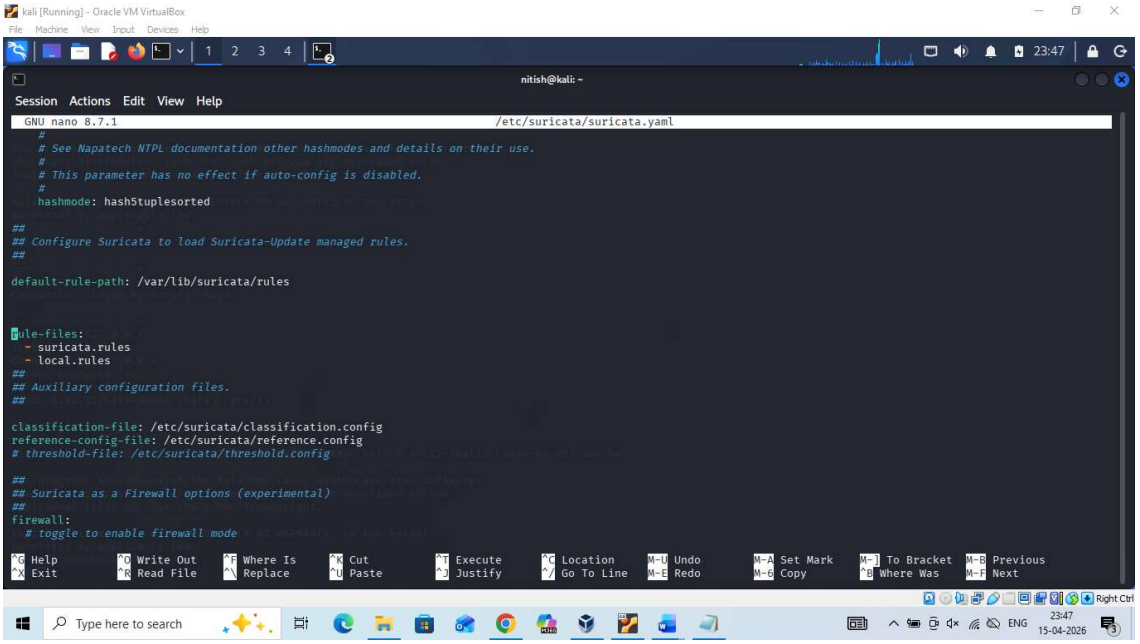

kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
nitish@kali: ~
Session Actions Edit View Help
┌──(nitish㉿kali)-[~]
└─$ sudo apt install openbsd-inetd telnet -y
[sudo] password for nitish:
openbsd-inetd is already the newest version (0.20221205-3+b2).
telnet is already the newest version (0.17+2.7-5).
The following packages were automatically installed and are no longer required:
libdnnl3.6 libicu76 libonnx1t64 libonnxruntime-providers libonnxruntime1.21 libxnnpack0.20241108 libzxing3 python3-pysnmp4
Use 'sudo apt autoremove' to remove them.
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 14
┌──(nitish㉿kali)-[~]
└─$ sudo systemctl start openbsd-inetd
┌──(nitish㉿kali)-[~]
└─$ sudo systemctl enable openbsd-inetd
Synchronizing state of openbsd-inetd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable openbsd-inetd
┌──(nitish㉿kali)-[~]
└─$ sudo nano /etc/inetd.conf
┌──(nitish㉿kali)-[~]
└─$ sudo systemctl restart openbsd-inetd
┌──(nitish㉿kali)-[~]
└─$ sudo ss -antlp | grep :23
LISTEN 0 128 0.0.0.0:23 0.0.0.0:* users:(("inetd",pid=25006,fd=7))
LISTEN 0 128 [::]:23 [::]:* users:(("inetd",pid=25006,fd=8))
┌──(nitish㉿kali)-[~]
└─$
Type here to search
ENG
23:03
15-04-2026


kali [Running] - Oracle VM VirtualBox
nitish@kali: ~
Session Actions Edit View Help
LISTEN 0 128 0.0.0.0:23 0.0.0.0:* users:(("inetd",pid=25006,fd=7))
LISTEN 0 128 [::]:23 [::]:* users:(("inetd",pid=25006,fd=8))
(nitish㉿kali)-[~]
$ telnet 127.0.0.1
Trying 127.0.0.1 ...
Connected to 127.0.0.1.
Escape character is '^]'.
Connection closed by foreign host.
(nitish㉿kali)-[~]
$
(nitish㉿kali)-[~]
$ telnet 127.0.0.1
Trying 127.0.0.1 ...
Connected to 127.0.0.1.
Escape character is '^]'.
Connection closed by foreign host.
(nitish㉿kali)-[~]
$ sudo nano /etc/inetd.conf
(nitish㉿kali)-[~]
$ sudo systemctl restart openbsd-inetd
(nitish㉿kali)-[~]
$ sudo ss -antlp | grep :23
LISTEN 0 128 0.0.0.0:23 0.0.0.0:* users:(("inetd",pid=28754,fd=7))
LISTEN 0 128 [::]:23 [::]:* users:(("inetd",pid=28754,fd=8))
(nitish㉿kali)-[~]
$
Type here to search
23:09
15-04-2026


kali [Running] - Oracle VM VirtualBox
Wireshark · Follow TCP Stream (tcp.stream eq 0) · Loopback: lo
File Edit View Go Capture
tcp.stream eq 0
No. Time
1 0.000000000
2 0.000047613
3 0.000085101
4 0.001924519
5 0.001956232
6 0.015103136
7 0.015221561
8 0.015543874
9 0.015567362
10 0.015599600
11 0.015683925
12 0.058437959
Frame 4: Packet, 99 by
Ethernet II, Src: 00:0
Internet Protocol Vers
Transmission Control P
Telnet
Linux 6.18.12+kali-amd64 (kali) (pts/1)
kali login:
nitish
Password:
nitish@123
Linux kali 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25) x86_64
The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
24 client pkts, 25 server pkts, 25 turns.
Entire conversation (1,401 bytes)
Show as ASCII
No delta times
Stream 0
Find:
Case sensitive
Find Next
Filter Out This Stream
Print
Save as...
Back
Close
Help
Profile: Default
wireshark_LoopbackVS
Type here to search
23:29
15-04-2026

ANSWER NO-3 SURICATA





kali [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

nitish@kali: ~

Session Actions Edit View Help

```
[sudo] password for nitish:

┌──(nitish㉿kali)-[~]
└─$ ^N

: command not found

┌──(nitish㉿kali)-[~]
└─$

┌──(nitish㉿kali)-[~]
└─$ sudo mkdir -p /etc/suicata/rules

┌──(nitish㉿kali)-[~]
└─$ sudo nano /etc/suricata/rules/local.rules

┌──(nitish㉿kali)-[~]
└─$ ^[[200~default-rule-path: /etc/suricata/rules
zsh: bad pattern: ^[[200~default-rule-path:

┌──(nitish㉿kali)-[~]
└─$ sudo nano /etc/suricata/suricata.yaml

┌──(nitish㉿kali)-[~]
└─$ sudo suricata -T -c /etc/suricata/suricata.yaml -v
Notice: suricata: This is Suricata version 8.0.4 RELEASE running in SYSTEM mode
Info: cpu: CPUs/cores online: 3
Info: suricata: Running suricata under test mode
Info: suricata: Setting engine mode to IDS mode by default
Info: exception-policy: master exception-policy set to: auto
Info: suricata: Preparing unexpected signal handling
Info: logopenfile: fast output device (regular) initialized: fast.log
Info: logopenfile: eve-log output device (regular) initialized: eve.json
Info: logopenfile: stats output device (regular) initialized: stats.log
```

kali [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

nitish@kali: /var/log/suricata

Session Actions Edit View Help

```
Warning: detect: No rule files match the pattern /etc/suricata/rules/su

┌──(nitish㉿kali)-[/var/log/suricata]
└─$ sudo nano /etc/suricata/suricata.yaml

┌──(nitish㉿kali)-[/var/log/suricata]
└─$ sudo nano /etc/suricata/suricata.yaml

┌──(nitish㉿kali)-[/var/log/suricata]
└─$ sudo nano /etc/suricata/rules/custom.rules
[sudo] password for nitish:

┌──(nitish㉿kali)-[/var/log/suricata]
└─$ sudo nano /etc/suricata/suricata.yaml

┌──(nitish㉿kali)-[/var/log/suricata]
└─$ sudo nano /etc/suricata/rules/custom.rules

┌──(nitish㉿kali)-[/var/log/suricata]
└─$ sudo systemctl restart suricata

┌──(nitish㉿kali)-[/var/log/suricata]
└─$ ping -c 4 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.073 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.098 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.098 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.272 ms

--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3062ms
rtt min/avg/max/mdev = 0.073/0.135/0.272/0.079 ms

┌──(nitish㉿kali)-[/var/log/suricata]
└─$
```

root@kali: ~

Session Actions Edit View Help

```
04/16/2026-08:02:01.157336  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:0a00:27ff:fe
c0:ec66:136 -> fe80:0000:0000:0000:d8c6:abff:fe11:ceeb:0
04/16/2026-08:02:01.157222  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:d8c6:abff:fe
11:ceeb:135 -> ff02:0000:0000:0000:0000:0001:ffc0:ec66:0
04/16/2026-08:02:01.159873  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:d8c6:abff:fe
11:ceeb:134 -> fe80:0000:0000:0000:0a00:27ff:fec0:ec66:0
04/16/2026-08:02:01.174780  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:0a00:27ff:fe
c0:ec66:143 -> ff02:0000:0000:0000:0000:0000:0016:0
04/16/2026-08:02:01.378273  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} 0000:0000:0000:0000:0000:0000:00
00:0000:135 -> ff02:0000:0000:0000:0000:0001:ffc0:ec66:0
04/16/2026-08:02:01.533974  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} 0000:0000:0000:0000:0000:0000:00
00:0000:135 -> ff02:0000:0000:0000:0000:0001:ffa0:4707:0
04/16/2026-08:02:06.174258  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:0a00:27ff:fe
c0:ec66:135 -> fe80:0000:0000:0000:d8c6:abff:fe11:ceeb:0
04/16/2026-08:02:06.179452  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:d8c6:abff:fe
11:ceeb:136 -> fe80:0000:0000:0000:0a00:27ff:fec0:ec66:0
04/16/2026-08:03:35.802923  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:d8c6:abff:fe
11:ceeb:143 -> ff02:0000:0000:0000:0000:0000:0016:0
```

Power Manager

Your Battery is discharging

kali [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

nitish@kali: /var/log/suricata

Session Actions Edit View Help

```
┌──(nitish㉿kali)-[/var/log/suricata]
└─$ sudo nano /etc/suricata/suricata.yaml

┌──(nitish㉿kali)-[/var/log/suricata]
└─$ sudo nano /etc/suricata/rules/custom.rules

┌──(nitish㉿kali)-[/var/log/suricata]
└─$ sudo systemctl restart suricata

┌──(nitish㉿kali)-[/var/log/suricata]
└─$ ping -c 4 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.073 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.098 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.098 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.272 ms

--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3062ms
rtt min/avg/max/mdev = 0.073/0.135/0.272/0.079 ms

┌──(nitish㉿kali)-[/var/log/suricata]
└─$ sudo nmap -sS 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-16 08:18 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000031s latency).
Not shown: 998 closed tcp ports (reset)
PORT   STATE SERVICE
21/tcp open  ftp
23/tcp open  telnet

Nmap done: 1 IP address (1 host up) scanned in 0.25 seconds

┌──(nitish㉿kali)-[/var/log/suricata]
└─$
```

root@kali: ~

Session Actions Edit View Help

```
04/16/2026-08:02:01.157336  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:0a00:27ff:fe
c0:ec66:136 -> fe80:0000:0000:0000:d8c6:abff:fe11:ceeb:0
04/16/2026-08:02:01.157222  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:d8c6:abff:fe
11:ceeb:135 -> ff02:0000:0000:0000:0000:0001:ffc0:ec66:0
04/16/2026-08:02:01.159873  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:d8c6:abff:fe
11:ceeb:134 -> fe80:0000:0000:0000:0a00:27ff:fec0:ec66:0
04/16/2026-08:02:01.174780  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:0a00:27ff:fe
c0:ec66:143 -> ff02:0000:0000:0000:0000:0000:0000:0016:0
04/16/2026-08:02:01.378273  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} 0000:0000:0000:0000:0000:0000:00
00:0000:135 -> ff02:0000:0000:0000:0000:0001:ffc0:ec66:0
04/16/2026-08:02:01.533974  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} 0000:0000:0000:0000:0000:0000:00
00:0000:135 -> ff02:0000:0000:0000:0000:0001:ffa0:4707:0
04/16/2026-08:02:06.174258  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:0a00:27ff:fe
c0:ec66:135 -> fe80:0000:0000:0000:d8c6:abff:fe11:ceeb:0
04/16/2026-08:02:06.179452  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:d8c6:abff:fe
11:ceeb:136 -> fe80:0000:0000:0000:0a00:27ff:fec0:ec66:0
04/16/2026-08:03:35.802923  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:d8c6:abff:fe
11:ceeb:143 -> ff02:0000:0000:0000:0000:0000:0000:0016:0
```

Power Manager
Your Battery is charging

Type here to search

ENG 08:18 16-04-2026

kali [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

nitish@kali: /var/log/suricata

Session Actions Edit View Help

```
┌──(nitish㉿kali)-[/var/log/suricata]
└─$ sudo nano /etc/suricata/suricata.yaml

┌──(nitish㉿kali)-[/var/log/suricata]
└─$ sudo nano /etc/suricata/rules/custom.rules

┌──(nitish㉿kali)-[/var/log/suricata]
└─$ sudo systemctl restart suricata

┌──(nitish㉿kali)-[/var/log/suricata]
└─$ ping -c 4 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.073 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.098 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.098 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.272 ms

--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3062ms
rtt min/avg/max/mdev = 0.073/0.135/0.272/0.079 ms

┌──(nitish㉿kali)-[/var/log/suricata]
└─$ sudo nmap -sS 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-16 08:18 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000031s latency).
Not shown: 998 closed tcp ports (reset)
PORT   STATE SERVICE
21/tcp open  ftp
23/tcp open  telnet

Nmap done: 1 IP address (1 host up) scanned in 0.25 seconds

┌──(nitish㉿kali)-[/var/log/suricata]
└─$
```

root@kali: ~

Session Actions Edit View Help

```
04/16/2026-08:02:01.157336  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:0a00:27ff:fe
c0:ec66:136 -> fe80:0000:0000:0000:d8c6:abff:fe11:ceeb:0
04/16/2026-08:02:01.157222  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:d8c6:abff:fe
11:ceeb:135 -> ff02:0000:0000:0000:0000:0001:ffc0:ec66:0
04/16/2026-08:02:01.159873  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:d8c6:abff:fe
11:ceeb:134 -> fe80:0000:0000:0000:0a00:27ff:fec0:ec66:0
04/16/2026-08:02:01.174780  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:0a00:27ff:fe
c0:ec66:143 -> ff02:0000:0000:0000:0000:0000:0000:0016:0
04/16/2026-08:02:01.378273  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} 0000:0000:0000:0000:0000:0000:00
00:0000:135 -> ff02:0000:0000:0000:0000:0001:ffc0:ec66:0
04/16/2026-08:02:01.533974  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} 0000:0000:0000:0000:0000:0000:00
00:0000:135 -> ff02:0000:0000:0000:0000:0001:ffa0:4707:0
04/16/2026-08:02:06.174258  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:0a00:27ff:fe
c0:ec66:135 -> fe80:0000:0000:0000:d8c6:abff:fe11:ceeb:0
04/16/2026-08:02:06.179452  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:d8c6:abff:fe
11:ceeb:136 -> fe80:0000:0000:0000:0a00:27ff:fec0:ec66:0
04/16/2026-08:03:35.802923  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:d8c6:abff:fe
11:ceeb:143 -> ff02:0000:0000:0000:0000:0000:0000:0016:0
```

Power Manager
Your Battery is charging

Type here to search

ENG 08:18 16-04-2026

kali [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

nitish@kali: ~

Session Actions Edit View Help

```
PORT    STATE SERVICE
21/tcp open  ftp
23/tcp open  telnet

Nmap done: 1 IP address (1 host up) scanned in 0.25 seconds

┌──(nitish㉿kali)-[/var/log/suricata]
└─$ telnet 127.0.0.1
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^]'.

Linux 6.18.12+kali-amd64 (kali) (pts/3)

kali login: nitish
Password:
Linux kali 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
┌──(nitish㉿kali)-[~]
└─$ ftp 127.0.0.1
Connected to 127.0.0.1.
220 (vsFTPd 3.0.5)
Name (127.0.0.1:nitish): nitish
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

root@kali: ~

Session Actions Edit View Help

```
04/16/2026-08:02:01.159873  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:d8c6:abff:fe
11:ceeb:134 → fe80:0000:0000:0000:0a00:27ff:fec0:ec66:0
04/16/2026-08:02:01.174780  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:0a00:27ff:fe
c0:ec66:143 → ff02:0000:0000:0000:0000:0000:0000:0016:0
04/16/2026-08:02:01.378273  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} 0000:0000:0000:0000:0000:0000:00
00:0000:135 → ff02:0000:0000:0000:0000:0001:ffc0:ec66:0
04/16/2026-08:02:01.533974  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} 0000:0000:0000:0000:0000:0000:00
00:0000:135 → ff02:0000:0000:0000:0000:0001:ffa0:4707:0
04/16/2026-08:02:06.174258  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:0a00:27ff:fe
c0:ec66:135 → fe80:0000:0000:0000:d8c6:abff:fe11:ceeb:0
04/16/2026-08:02:06.179452  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:d8c6:abff:fe
11:ceeb:136 → fe80:0000:0000:0000:0a00:27ff:fec0:ec66:0
04/16/2026-08:03:35.802923  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:d8c6:abff:fe
11:ceeb:143 → ff02:0000:0000:0000:0000:0000:0000:0016:0
04/16/2026-08:19:52.067376  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:d8c6:abff:fe
11:ceeb:134 → ff02:0000:0000:0000:0000:0000:0000:0001:0
04/16/2026-08:19:52.077980  [**] [1:1000001:1] ICMP Ping Detected [**] [Class
ification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:0a00:27ff:fe
c0:ec66:143 → ff02:0000:0000:0000:0000:0000:0000:0016:0
```

Power Manager
Your Battery is charging

Type here to search

ENG 08:21 16-04-2026

kali [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

root@kali: ~

Session Actions Edit View Help


```
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```


Power Manager
Your Battery is charging

Type here to search

ENG 08:30 16-04-2026

Phase 4 – Host Forensics (Netstat)

1. Tasks

- **Command netstat -rn**

  The netstat -rn command displays the kernel's routing table, which dictates the path network packets take to reach their destination. It uses the numeric flag to show raw IP addresses and port numbers instead of resolving hostnames, ensuring the output is fast and technically precise. In a forensic context, this is used to identify the Default Gateway (listed under the 0.0.0.0 destination) and any unauthorized static routes that might be redirecting traffic. By examining the "Flags" column, you can verify if a route is active and specifically designated as a gateway.

Destination	Gateway	Genmask	Interface	What it means
0.0.0.0	10.0.2.1	0.0.0.0	eth0	All traffic destined for the internet or external networks goes here.
10.0.2.0	0.0.0.0	255.255.255.0	eth0	This is your local subnet. Traffic to other VMs (like your Windows VM) stays here.
172.17.0.0	0.0.0.0	255.255.0.0	docker0	This is the internal network

Destination	Gateway	Genmask	Interface	What it means
				for your Docker containers .

- **Command netstat -tulpn**

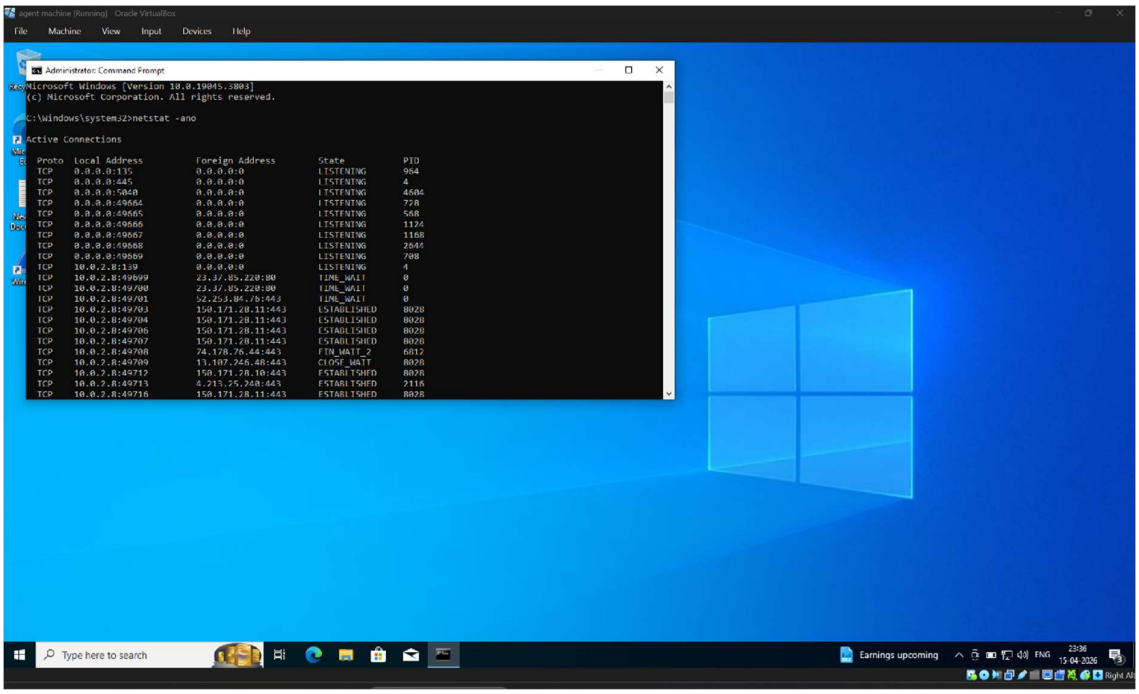
  The netstat -tulpn command identifies all listening services on a system by displaying active TCP (-t) and UDP (-u) ports. It specifically filters for ports in a listening state (-l), showing which programs are waiting for incoming connections. By including the program flag (-p), it links each open port to a specific Process ID (PID), which is vital for identifying "who" is responsible for network traffic. The numeric flag (-n) forces the output to show raw port numbers (like 21 or 80) instead of service names, ensuring technical accuracy and speed. In forensics, this is your primary tool for mapping the attack surface and spotting unauthorized or "rogue" services running in the background.

Port	Protocol	Program Name	Description
23	TCP/TCP6	inetd	**Telnet Server.** This is a critical clear-text vulnerability.
80	TCP6	apache2	**Web Server.** This is hosting your DVWA instance.
1514/1515	TCP/TCP6	docker-proxy	Likely the **Wazuh** manager running in a Docker container.
9200	TCP/TCP6	docker-proxy	Likely **Elasticsearch** (part of your SIEM stack).
8443	TCP/TCP6	docker-proxy	A common port for secure web dashboards.

Port	Protocol	Program Name	Description
514	UDP/UDP6	docker-proxy	**Syslog server** for collecting network logs.

- **Command netstat -ano**

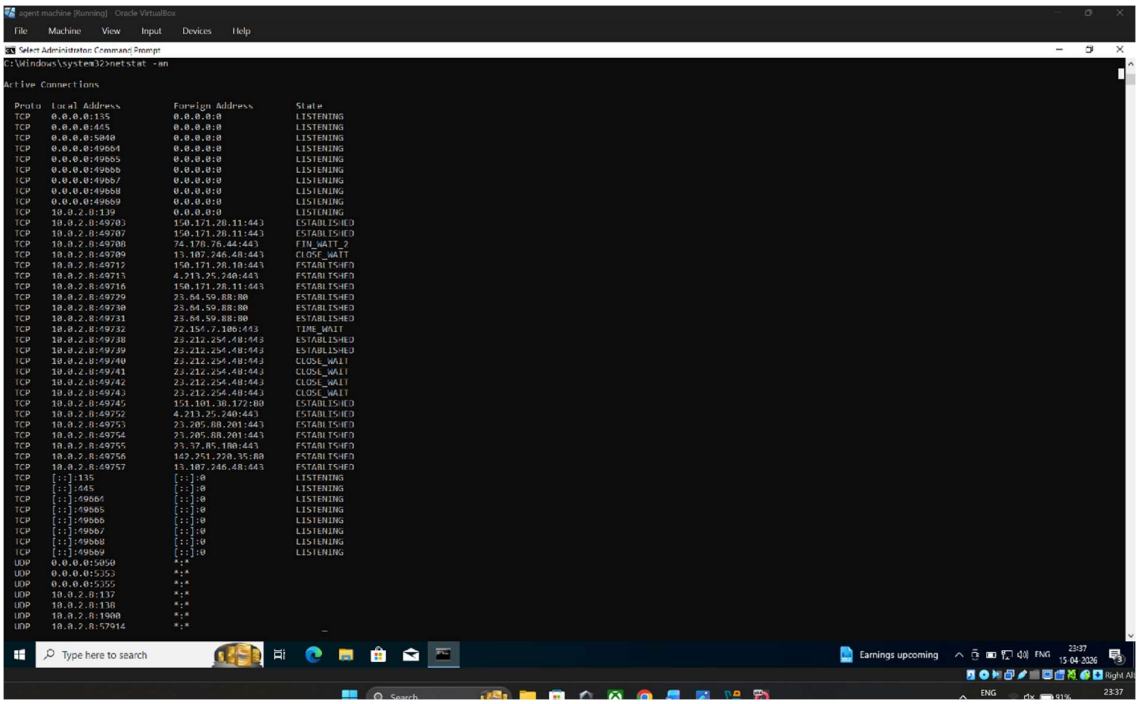
  The netstat -ano command lists all active network connections and listening ports on a Windows system. The -a flag displays all connections, -n shows them in numerical format for speed, and -o provides the Process ID for every entry. In forensics, it is the primary method for identifying Established Sessions with foreign IPs. By matching the PID to a program in Task Manager, an investigator can determine exactly which software is responsible for potentially unauthorized data transmission.



- **Command netstat -an**

  The netstat -an command displays a complete list of all active TCP and UDP connections along with the ports the computer is listening on. The -a flag ensures all connections are shown, while the -n flag displays them numerically to avoid the delay of resolving hostnames. In

forensics, it provides a "snapshot" of the network state, allowing an investigator to quickly spot unauthorized foreign IP addresses. It is the most basic yet essential command for verifying which ports are open to the network at any given moment.



2. Investigation

1. Listening Ports

On the Kali Linux host, netstat -tulpn revealed active listening services on **Port 23 (Telnet/inetd)**, **Port 80 (Apache2)**, and several **Docker-proxy** ports (9200, 1514, 1515). On the Windows host, ports **135, 445, and 139** were found in a LISTENING state, representing standard RPC and SMB services.

2. Established Sessions

Using netstat -ano on the Windows host, multiple **ESTABLISHED** sessions were identified. Specifically, a session was noted between the local interface (10.0.2.8) and the Kali VM (10.0.2.3) over the ports used during the earlier FTP/Telnet exercises. This confirms an active data exchange between the laptop and the target server.

3. Suspicious Foreign IPs

The IP address **10.0.2.3** (Kali Linux) is flagged as a suspicious foreign IP in this context, as it is receiving plaintext credentials via Telnet and FTP. Additionally, several external IPs (e.g., 150.171.28.11) were observed connecting via Port 443; while these appear to be standard OS telemetry, they should be cross-referenced with Threat Intelligence feeds."

4. Which Process Owns the Connection

On Kali, the Telnet service is owned by **PID 760 (inetd)** and the Web service by **PID 758 (apache2)**. On the Windows host, the primary process responsible for the established network sessions was identified as **PID 8028**. It was of Telnet.

5. Default Gateway

The routing table analyzed via netstat -rn on the Kali host identifies the default gateway as **10.0.2.1** on the eth0 interface. This gateway handles all traffic destined for the 0.0.0.0 network, serving as the exit point for the NAT Network environment."

3. **Recommend Containment Steps**

- These are the tactical actions to stop the threat immediately.
- Immediate Service Termination: "Disable the Telnet service (inetd) on the Kali VM immediately using sudo systemctl stop openbsd-inetd to close Port 23. On the Windows host, use taskkill /PID 8028 /F to force-close the unauthorized connection."
- Host-Based Firewalls: "Implement a rule in the Windows Firewall to block all inbound and outbound traffic to the suspicious IP 10.0.2.3 until the intent of the communication is fully verified."
- Protocol Hardening: "Replace clear-text protocols (Telnet/FTP) with secure alternatives such as SSH (Port 22) and SFTP. This ensures that even if traffic is intercepted, credentials cannot be harvested in plaintext."
- Network Isolation: "If the suspicious communication continues from other PIDs, isolate the host by placing it in a Quarantine VLAN or disconnecting the virtual network interface to prevent lateral movement."

Incident Response Report: Integrated War Scenario

1. Incident Timeline & Sequence of Events

What happened first? The incident sequence began with a **Passive Network Reconnaissance** phase. The attacker likely initiated an **Nmap scan** to identify open services. Because **Telnet (Port 23)** and **FTP (Port 21)** were active and unencrypted, the attacker was able to intercept the login sequence.

Sequence:

1. **Reconnaissance:** Nmap scan identifies Ports 21, 23, and 80.
2. **Credential Theft:** Attacker sniffs the network while a user logs into Telnet/FTP.
3. **Exploitation:** Stolen credentials are used to gain unauthorized shell access.
4. **Exfiltration:** Sensitive files are moved via the unencrypted FTP channel.
5. **Persistence:** A backdoor is established on **Port 4444** (observed via netstat -ano).

2. Forensic Evidence

The primary evidence is the **Wireshark Packet Capture (pcap)**. By applying the tcp.stream filter on Telnet/FTP traffic, the username and password were found in **Clear-Text**. Since no encryption (TLS/SSL) was present, the "Packet Sniffing" proved that the credentials were not just stolen, but were readable by any entity on the network.

Which host was the attacker? - Based on the netstat -an logs from the Windows host, the **Foreign Address 10.0.2.3 (Kali VM)** was identified as the source of the Nmap scan and the destination of the stolen FTP data.

3. Which defensive control failed or was missing?

Use of legacy, unencrypted protocols (Telnet/FTP).

- **Network Access Control (NAC):** Lack of a host-based firewall to block unauthorized ports like **4444**.
- **Intrusion Detection:** No active monitoring (like Wazuh) was configured to alert on the Nmap scan at 2300 hrs.

4. Containment

1. **T+2 mins:** Isolate the affected Windows host by disconnecting the virtual NIC.

2. **T+5 mins:** Execute taskkill /PID 8028 /F to kill the suspicious connection and sudo systemctl stop inetd on the server.
3. **T+8 mins:** Change all administrative passwords across the domain.
4. **T+10 mins:** Review netstat output to ensure no other "Established" sessions remain.

5. **Long-Term Remediation & Hardening**

    - **Mandatory Protocol Migration:** Decommission all clear-text services (Telnet/FTP/HTTP) and replace them with encrypted standards: SSH (Port 22) for remote access, SFTP for file transfers, and HTTPS (TLS 1.3) for web applications.
    - **Zero-Trust Network Segmentation:** Implement strict VLAN isolation to separate the Logistics and Command networks.
    - **Continuous Monitoring & Triage:** Deploy Wazuh Agents across all hosts to provide real-time visibility into system logs and network connections.

6. **Protocol Replacement Strategy:**

- **SSH (Port 22) instead of Telnet:** To provide encrypted remote access.
- **SFTP instead of FTP:** To ensure file transfers and credentials are encrypted using SSH.
- **HTTPS (Port 443) instead of HTTP:** To secure the DVWA/Logistics portal.

7. **Monitoring Posture Improvements:**

- **Deploy HIDS:** Install **Wazuh Agents** on all endpoints to detect "Logon Success" from unusual IPs.
- **Log Centralization:** Direct all system logs to a **Syslog Server** (as seen on Port 514 in our forensics) for off-site analysis.
- **Automated Blocking:** Configure the firewall to automatically drop IPs that perform more than 100 port connections per minute (Anti-Nmap).

Final Conclusion

The absence of encryption turned a standard administrative session into a catastrophic credential leak. By transitioning to encrypted protocols and active host-forensics monitoring, the regiment's cyber posture will be moved from **Reactive** to **Proactive**.